

1. Write down the postulates of quantum mechanics
2. Write down the differences between classical mechanics and quantum mechanics.
3. Explain the physical meanings of the postulates of quantum mechanics.
4. What do you mean by the term quantum system?
5. Define Hilbert space.
6. Define inner-product of two functions.
7. Define normalization, orthogonal, orthonormal, and completeness of functions.
8. Define the expectation value of an operator in inner-product.
9. What is an Hermitian operator? Show that the momentum operator is Hermitian.
10. Let us given an operator $\hat{Q} = i \frac{d}{d\varphi}$ where φ is the usual polar coordinate in two dimensions. (a) Is Q Hermitian? (b) Find out the eigenvalues and eigenfunctions of Q.
11. Write down the properties of eigenfunctions of Hermitian operators. Proof them.